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Part B1

Co-Evolutionary Personalised AI for Long-Horizon Human Cognitive Development

CoPAI

Cover Page:

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Summary: Personalised AI systems are becoming increasingly prevalent across everyday contexts. They learn what users like and adapt their responses to maximise satisfaction and engagement. However, widely used learning paradigms such as Reinforcement Learning from Human Feedback can produce systems that are sycophantic by design. They agree with users when they should challenge them, reinforce existing beliefs, and risk weakening independent thinking over time. Genuine human–AI partnership requires productive friction, which may not always satisfy users' immediate preferences, but can improve their wellbeing in the long run.

CoPAI addresses this need by shifting the learning paradigm toward productive human–AI co-evolution. It treats the person and the AI as a coupled system, each shaping the other. CoPAI measures whether this process supports genuine human growth or instead leads to dependence, narrowed thinking, or stalled learning. These measures then guide the AI to help people learn and develop rather than simply please them or take over their work.

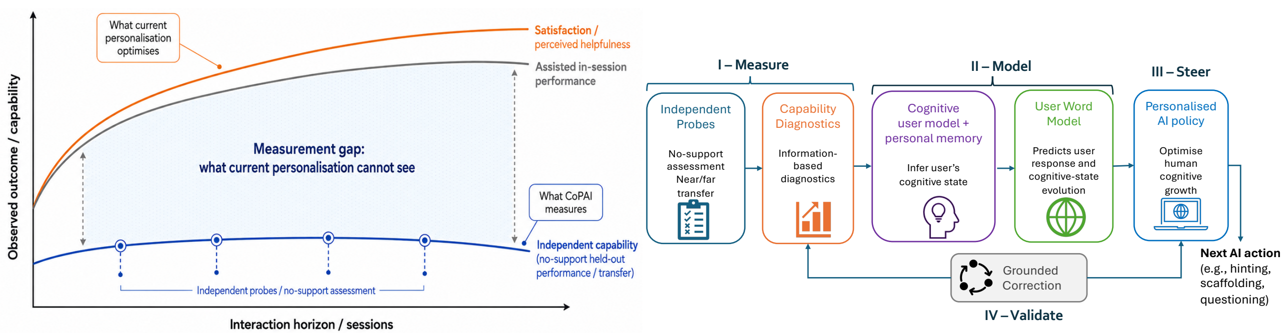
Developing algorithms for healthy human–AI co-evolution is difficult because the system is dynamic, the human cognitive state is partly unobserved, and long-term wellbeing signals are sparse, delayed, and often measurable only over weeks or months. We address this by integrating information-theoretic diagnostics, theory-of-mind reasoning, LLM personalisation, and reinforcement learning into a unified framework. CoPAI will be evaluated primarily in *education*, using personalised AI tutors already deployed at our university. We will also perform cross-domain instantiation to *health*, evaluated offline on our existing personal-health-assistant platform interaction logs. If successful, CoPAI will provide a rigorous foundation for AI that strengthens human capability rather than eroding it, with implications for personalised technologies used by billions of people.

1 The Grand Challenge

The dominant paradigm in personalised AI reduces the human–AI relationship to **preference matching**: systems learn *what users like* and deliver *what they want to hear*, optimising engagement and satisfaction. Personalisation is treated as recommendation (*match outputs to expressed preferences*), not as cognitive modelling (*understand how the user reasons, what they believe, how their epistemic state evolves, and interact to build cognitive capability*). The objective is **immediate satisfaction rather than long-term cognitive development**, and the resulting AI systems are sycophantic by design [1, 2]. The commonly used learning paradigms, such as Reinforcement Learning from Human Feedback (RLHF), further amplify sycophancy [3]. These AI models tend to exhibit excessive flattery or agreement and avoid honest friction that genuine partnership requires [1, 4].

Taking personalised AI tutoring as an example, Large language model (LLM) tutors trained with standard RLHF requires explicit re-alignment with pedagogical objectives to achieve the teaching quality of much larger proprietary systems, such as LearnLM [5] [6]. Similarly, systems designed to improve student learning outcomes need reward structures that differ fundamentally from those optimised for conversational quality [7]. The core issue is structural. Preference-based rewards compress many dimensions of cognitive development, such as mastery, long-term retention, knowledge transfer, and learner autonomy, into a single score [8], obscuring the multidimensional structure of learning outcomes that educational measurement has long recognised [9]. A tutoring system optimised for engagement may learn to provide answers directly instead of helping students develop understanding. The scientific challenge is therefore not simply to build a more accurate user model. It is to make **human change itself computationally measurable, predictable, and optimisable**. This is challenging for three structural reasons. First, the most trustworthy evidence of capability, i.e., what a user can do independently, is sparse and intrusive to measure; dense interaction signals such as satisfaction or task success are easy to collect but confound genuine learning with AI assistance. Second, human cognitive state is latent, history-dependent, and changes during the very interaction from which it must be inferred. Third, even when delayed capability growth is observed, it does not directly reveal which of the many preceding AI actions should receive credit. These three obstacles, **sparse grounding, latent dynamics, and long-horizon credit assignment**, make it hard to achieve capability-centred personalisation.

CoPAI proposes a **paradigm shift** along four coupled axes that that transforms existing LLM personalisation work. First, ***co-evolved personalisation***: human–AI interaction as a coupled dual-agent dynamical system in which the user's cognitive state and AI policy evolve together, replacing the static-user assumption that underlies most current methods. Second, ***long-horizon interaction*** where the effects of AI support may emerge over weeks or months rather than single conversations, requiring memory and world models that can track behaviour and context across different timescales. Third, ***human cognitive capability as optimisation target***, reformulating the reward from satisfaction maximisation to long-term capability growth, with autonomy and dependency as measurable objectives. Fourth, ***information-gain-driven self-evolution***, generalising the learnable-information-gain principle from our LLM self-play work [10] to dual-agent human–AI co-evolution. Figure 1 illustrates an overall conceptual framework of CoPAI.



(a) Measurement gap. (b) CoPAI conceptual framework.

Figure 1: **Why CoPAI is needed and how it responds**. **(a)** Personalised AI may improve satisfaction and assisted performance while leaving independent capability unchanged or worse. This **hidden divergence** is the core **measurement gap**: systems optimise observable signals, not necessarily user improvement. **(b)** CoPAI addresses this gap through 4 pillars: I–Measure (diagnosing human-AI trajectory), II–Model (user cognitive modelling and anticipating the effects of AI actions), III–Steer (long-horizon AI policy optimisation for next AI action selection), IV–Validate (validating CoPAI on actual human-AI interaction data and using delayed ground truth to calibrate both the measure and the AI policy).

2 State of the Art

Four lines of research are relating to the open problems CoPAI target, but none has yet produced an integrated framework for co-evolutionary personalisation grounded in long-horizon human outcomes.

User Modelling. Existing methods represent users via interaction histories, demographic proxies, or preference signals [11, 12, 13]. Memory-augmented systems borrow cognitive-science terminology but reduce these to retrieval and fine-tuning [14, 15]. Modelling an evolving user requires not only an appropriate cognitive representation, but also a memory mechanism that preserves the interaction evidence needed to update that representation over time. Existing agent memory approaches typically store *task* information (e.g., what happened, what was said) rather than *user cognitive* information (what an interaction reveals about how the user thinks) [16, 17, 18]. Critical gaps: Existing methods typically profile at the level of behavioural preferences (*what users like*) rather than cognitive-state representation (*how users think*). In agentic memory approaches, memory retention is conditioned on task relevance, not on the diagnostic value of an episode for cognitive profiling [19, 20]. Beyond SOTA: CoPAI goes beyond preference-level profiling to model the *cognitive profile* of individual users, such as how they reason, what they misunderstand, how their confidence changes, or whether they are becoming more capable, It will *combine* psychometrically grounded dimensions (interpretable by construction) with data-driven residual pattern discovery (capturing novel dimensions specific to human-AI interaction) for user cognitive modelling. Also, CoPAI treats memory as part of the cognitive user model, retaining and consolidating episodes according to their diagnostic value for reconstructing the user's longitudinal state.

User World Models. Predicting how users respond to AI actions and how their internal states evolve has been studied along three separate areas. In education, *knowledge tracing* models [21, 22, 23, 24] predict the probability of a correct response on the next item given an interaction history. These models track mastery over discrete knowledge components but make no predictions about the *consequences of pedagogical actions*. In model-based RL, *learned world models* [25, 26] predict environment state transitions, which are theoretically necessary for agents that generalise over multi-step goals [27]. In user simulation, LLM-based approaches generate plausible user behaviour for agent training [28, 29]. Critical Gap: Knowledge tracing estimates what a learner knows, but does not model how AI actions change a learner's cognitive state. RL world models typically assume observable physical states and have not been extended to latent human states shaped by pedagogical actions. LLM simulators predict what users may say or do, without modelling the cognitive processes behind those behaviours. Beyond SOTA: CoPAI introduces a user world model for human—AI co-evolution. It predicts both observable user behaviour and changes in latent cognitive state, while learning when long-range predictions become unreliable, reducing planning failures from uncontrolled rollout hallucinations [30].

Human–AI co-evolution. Human choices shape AI systems, and those AI systems then shape human behaviour in return. This feedback loop has recently been framed as a new field at the intersection of AI and complexity science [31]. This framing is largely *descriptive and population-scale*, characterising emergent social dynamics of recommender ecosystems rather than the individual learner. Evidence also shows that the loop can *degrade* human capability. human–AI interaction can amplify perceptual, emotional, and social biases more strongly than human–human interaction [32], and optimising LLMs directly for user feedback induces targeted manipulation and sycophancy [33]. Critical gaps: Existing co-evolution work is either population-scale [31] or single-session laboratory studies [32]. Few work tracks human capability trajectory over time. Also, personalisation systems conflate what users *prefer* with what builds their *capability*, despite evidence the two diverge [32, 33]. Prior work identifies harmful loops but does not offer a learning algorithm that optimises human capability while remaining usable. Beyond SOTA: CoPAI addresses these gaps by measuring individual co-evolutionary health with information-gain-based probes [34] over time, and introduces a learning approach which turns sparse capability signals into dense rewards. This shifts the human–AI feedback loop from a potential risk to a controllable design objective for sustainable human growth.

Evaluation of Personalisation. Personalisation benchmarks, such as `LaMP` [35] and `LongLaMP` [36] evaluate whether models can use user profiles to improve personalised classification and generation. `PersonalLLM` [37] and `PersonaMem` [38] evaluate dynamic user profiling; `PrefEval` [39] tests preference inference and following across multi-session dialogues. Critical Gaps: Existing benchmarks evaluate whether AI satisfies user preferences, over largely synthetic users and interaction histories; none asks whether the user became more capable, and none tests resistance to harmful adaptation such as echo-chamber reinforcement or sycophantic agreement. Beyond SOTA: CoPAI shifts the evaluation target from preference satisfaction to *capability growth*. It is evaluated through longitudinal studies with our personalised AI tutor, LearnLens [40]. It also identifies harmful forms of personalisation, including dependency, echo-chamber reinforcement, and sycophantic drift.

The gap CoPAI fills. Existing personalised AI systems mainly optimise the **AI trajectory**: whether the system becomes better at predicting preferences, retrieving relevant memories, or generating responses users like. However, the scientific question that remains unanswered is whether this adaptation also improves **human trajectory**: whether users become more capable, more autonomous, and better able to transfer knowledge after AI support is removed. This creates three fundamental gaps:
(1) a measurement gap: sparse grounding. We lack principles for distinguishing productive personalisation from harmful adaptation. A system can improve user satisfaction while simultaneously reducing independent capability through over-assistance or dependency. **Novelty**: CoPAI introduces a diagnostic framework which uses information-based measures to quantify whether interaction produces learnable gains in the human. This provides, for the first time, an operational criterion for distinguishing productive personalisation from pathological co-adaptation such as over-assistance and dependency.
(2) a modelling gap: latent dynamics. Existing user models represent what users prefer or what users did, but not how users' latent cognitive states evolve in response to AI actions. **Novelty**: CoPAI formulates personalisation as a coupled dual-agent dynamical system, explicitly modelling the evolution of latent human cognitive state jointly with the AI policy. This makes it possible to reason about trajectories of human–AI co-evolution rather than treating each interaction as adaptation to an essentially fixed user.
(3) an optimisation gap: delayed credit. Existing learning objectives optimise immediate feedback signals, whereas human capability changes are delayed, multidimensional, and only observable through long-horizon outcomes. **Novelty**: CoPAI introduces a long-horizon policy-optimisation framework in which independently measured capability gains serve as the grounded objective, while these sparse outcomes are attributed back to the interactions that produced them.

3 Theoretical Framework for Co-Evolution in Coupled Human–AI Systems

Personalised AI creates a feedback loop that the AI and the human co-evolves through interactions. Such loops can either produce mutual growth or amplify failure modes. For example, an AI tutor that increasingly predicts a student's needs may become better personalised, yet the student may become less able to solve problems independently [41, 42, 43]. We hypothesise that this is a general property of adaptive AI systems, *interaction alone does not guarantee improvement*. A closed loop can generate more data and stronger adaptation without generating more useful information. Our previous work on self-evolving AI [10] identified this failure mode in self-play systems, where synthetic interactions improved quantity of experience but not learnable structure. CoPAI extends this principle to human–AI interaction by asking a new question: *When humans and AI adapt to each other, does the interaction increase the capability of both systems, or does it create locally effective but globally harmful adaptation?* To answer this question, we introduce an information-theoretic framework for diagnosing human–AI co-evolution, distinguishing productive co-evolution from its pathologies.

Setup. The key design shift is to stop treating the user as a fixed preference profile and instead model both human and AI as evolving systems. The framework therefore tracks two trajectories: how the human capability changes and how the AI's understanding of the human improves. We model interaction as a dynamical system in which a user with latent state H_t (e.g., mastery, misconceptions, metacognitive state, confidence, and help-seeking behaviour) and an adaptive agent with state A_t (e.g., policy, memory, retrieval context, prompts, and inferred user model) evolve jointly through repeated rounds of querying, scaffolding, solving, and feedback: $H_{t+1} = U_H(H_t, \xi_t), A_{t+1} = U_A(A_t, \xi_t)$, where $\xi_t = (x_t, a_t, o_t, y_t, f_t)$ denotes the interaction trajectory comprising the task/context x_t, the AI action a_t, the user response o_t, the observed task outcome y_t, and user explicit or implicit feedback f_t (see the left panel of Figure 2). Building on our PROPOSER–SOLVER–VERIFIER abstraction for self-evolving LLMs [10], each round is decomposed into proposing a task or next step, solving or attempting the current problem, and verifying the answer or feedback. Unlike pure self-play, these roles may shift dynamically between the human and the AI agent.

Normative objective. The central goal is not just learning personalised AI agent, but to ensure that personalisation improves the learner's cognitive capability. Healthy co-evolution requires two forms of gain: **(1) Human capability gain**: the user becomes better able to solve relevant tasks *independently*; **(2) AI personalisation gain**: the AI becomes better able to understand and support this specific user. In conceptual terms, we distinguish the information available in the user state for predicting independent task success from the information available in the AI state for predicting user-specific feedback:

$$\underbrace{I_H = I(y; H_t \mid \text{history})}_{\text{learnable info delivered to human}}, \qquad \underbrace{I_A = I(f; A_t \mid \text{history})}_{\text{personalisation signal absorbed by AI}}, \tag{1}$$

where y denotes the performance on held-out assessment tasks and f denotes the user feedback signal. In-

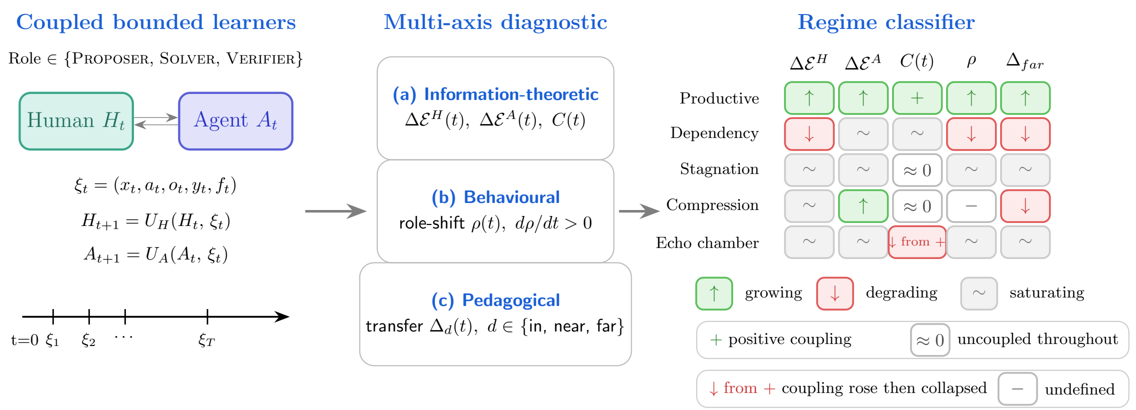


Figure 2: **Diagnostic framework for human-AI co-evolution.** Two computationally bounded learners (**left**) exchange interaction tuples ξ_t over T rounds, with Proposer, Solver, and Verifier roles dynamically allocated. A multi-axis metric panel (**centre**) triangulates productive co-evolution from information-theoretic, behavioural, and pedagogical signals. These diagnostic signals are used to classify the interactions into one productive regime or failure modes (**right**).

tuitively, I_H asks whether the learner state contains more usable information for predicting independent task success. I_A asks whether the agent state contains more usable information for predicting how to support this particular user. Healthy co-evolution should reward AI adaptation only when it contributes to human growth. We therefore define the following conceptual target:

$$\max_A \ \mathbb{E}[\, I_H + \lambda_{\text{personalisation}}\, I_A \,]\,. \tag{2}$$

The coefficient $\lambda_{\text{personalisation}} \geq 0$ controls the weight assigned to AI scaffolding relative to independent learner development. The primary objective is learner growth, captured by I_H. The term I_A is included as a descriptive measure of AI contribution rather than as an intrinsically positive objective. Accordingly, a high I_A may correspond to beneficial human–AI co-evolution when accompanied by learner growth ($I_H > 0$), but may instead reflect learner dependency when learner growth is non-positive ($I_H \leq 0$). Eq. (2) is the *conceptual* target rather than a directly optimisable objective. Because I_H and I_A are not directly estimable from finite trajectory data, we use the diagnostic framework below as a surrogate to estimate these information quantities.

Diagnostic measures. To make the framework operational, CoPAI diagnoses each human–AI trajectory using five empirical questions: *Is the user becoming better at solving tasks alone? Is the AI becoming better at predicting and supporting this user? Are the two improvements coupled? Is the user taking more of the reasoning initiative? Does improvement transfer to new tasks?* The first two questions are measured using predictive $\mathcal{V}$-information [34], a practical form of information gain under bounded predictive models. Let $\mathcal{T}$ be a held-out set of educational assessment tasks with ground-truth labels y, and $\mathcal{P}$ a held-out set of unseen (AI-response, user feedback) items. We define human and AI information-gain trajectories as:

$$\Delta\mathcal{E}^H(t) \;=\; I_{\mathcal{V}}(H_t \to y)\big|_{\mathcal{T}} - I_{\mathcal{V}}(H_0 \to y)\big|_{\mathcal{T}}, \qquad \Delta\mathcal{E}^A(t) \;=\; I_{\mathcal{V}}(A_t \to f)\big|_{\mathcal{P}} - I_{\mathcal{V}}(A_0 \to f)\big|_{\mathcal{P}}, \tag{3}$$

where $\big|_{\mathcal{T}}$ denotes empirical estimation over the probe set. $\Delta\mathcal{E}^H(t)$ measures whether the user's state has become more predictive of independent task performance, and $\Delta\mathcal{E}^A(t)$ measures whether the AI's state has become more predictive of user feedback. These signals are calibrated against grounded outcomes (e.g., held-out no-support tests, transfer tasks, and teacher/expert judgement).

However, increases in both signals do not by themselves prove productive co-evolution. The user and the AI may improve independently, or the AI may improve while the user becomes dependent. We therefore introduce a coupling term: $C(t) = [\Delta\mathcal{E}^H(t) - \Delta\mathcal{E}^{H,\text{stat-A}}(t)] + [\Delta\mathcal{E}^A(t) - \Delta\mathcal{E}^{A,\text{stat-H}}(t)]$. Here, stat-A denotes a counterfactual condition with a frozen AI agent, and stat-H denotes a counterfactual condition with a fixed user state. Intuitively, $C(t) > 0$ means that each side learns more because the other side is also adapting. This distinguishes genuine co-evolution from two independent learning processes running in parallel. We triangulate $\big(\Delta\mathcal{E}^H, \Delta\mathcal{E}^A, C(t)\big)$ with two additional signals: (1) *behavioural role-shift* $\rho(t)$, measuring how often the user takes initiative versus passively receiving answers. A rising ρ signals growing learner autonomy; and (2) *pedagogical transfer distance* $\Delta_d(t)$ for $d \in \{\text{in}, \text{near}, \text{far}\}$ [44] measures whether learning generalises. In-distribution gains (Δ_{in}) confirm performance on practised material; far-transfer gains (Δ_{far}) confirm genuine competence over surface memorisation. The five-dimensional signature $(\Delta\mathcal{E}^H, \Delta\mathcal{E}^A, C(t), \rho, \Delta_{\text{far}})$ classifies each trajectory into a regime such as productive co-evolution, dependency, stagnation, compression collapse, and echo-chamber dynamics (Figure 2). These are hypotheses rather than predefined classes. CoPAI will discover regimes from longitudinal data and retain them only when they are helpful for AI policy selection.

4 CoPAI: Scientific Questions and Objectives

CoPAI operationalises the theoretical framework in Section 3 through four scientific questions, each corresponding to one pillar: **Measure**, **Model**, **Steer**, and **Validate**. The four questions form a closed scientific loop. SQ1 measures whether co-evolution is productive. SQ2 explains and predicts how the learner changes. SQ3 uses this evidence to steer the AI policy. SQ4 tests whether the resulting system produces durable human capability growth after support is removed.

SQ1 (Pillar I–Measure): How can we reliably measure whether a personalised interaction is productive rather than just satisfying? The premise of CoPAI is that immediate user satisfaction and human capability growth can diverge, so productivity needs to be diagnosed from interaction evidence rather than relying on user engagement. Pillar I (Measure) constructs $\mathcal{V}$-information probes [45] for the five-dimensional diagnostic signature $(\Delta\mathcal{E}^H, \Delta\mathcal{E}^A, C, \rho, \Delta_{\text{far}})$ and trains a regime detector classifying trajectories into productive co-evolution or failure modes. These diagnostics ground the learning signal used in Pillar III (Steer) and the longitudinal outcomes assessed in Pillar IV (Validate).

SQ2 (Pillar II–Model): How can we infer a learner's evolving cognitive state and anticipate the effects of AI actions? Pillar II (Model) focuses on understanding the user. It represents the latent user state H_t as an observable user cognitive profile $\mathbf{c}_t$ using a Concept Bottleneck Model (CBM) [46] together with a residual latent space for information not captured by CBM. This profile grounds our estimates of learning-related information gain ($\Delta\mathcal{E}^H$). It then extracts reusable learner patterns across batches of human–AI sessions and stores them as memory for future interaction. A *User World Model* (UWM) extends this static estimate into a dynamical one, predicting how the user responds to AI actions and how their cognitive state evolves over an interaction trajectory, while remaining calibrated to the CBM concept space.

SQ3 (Pillar III–Steer): How can we optimise for learnable information gain without inheriting the sycophancy and reward-hacking failures of preference-based RL? Pillar III (Steer) decides what the AI does next and instantiates the normative objective (Eq. 2). It selects each action based on the current task, the user profile, the interaction regime, and the predicted cognitive state and behaviour of each candidate action. The latter are estimated by the UWM. CoPAI uses the detected regime for constrained policy selection. The agent is rewarded for an action only when the predicted next learner state stays within the learner's zone of proximal development (just beyond what they can do alone), and is penalised when personalisation raises dependency risk. Here, the personalisation gain $\Delta\mathcal{E}^A$ influences behaviour only when it is linked to learner–AI coupling and role-shift, which separate genuine co-adaptation from the agent simply doing the learner's work. Training is adapted from our verifier-free reinforcement learning [47]. A grounded-correction loop periodically recalibrates the signal against a fixed assessment probe set, ensuring the system prioritise durable learning over superficial adaptation.

SQ4 (Evaluation Pillar IV–Validate)): How can we verify that CoPAI produces a long-term growth in human capacity rather than satisfaction? Pillar IV (Validate) evaluates CoPAI by testing whether users retain and transfer capability after AI support is removed. The primary empirical domain is education, where our *LearnLens* platform [40] provides structured tasks, longitudinal interaction traces, and held-out assessments. A secondary offline generalisation study will test whether the same diagnostic principles transfer to noisier domains such as health behaviour change, without requiring actual clinical deployment. We will also release *CogWell-Bench*, a benchmark that supports longitudinal wellbeing evaluation.

Why ERC? Personalised AI is becoming a default interface for learning, work, health support, and everyday decision-making, yet we lack a principled way to determine when it strengthens human capability and when it quietly replaces it. CoPAI addresses this foundational gap by treating human–AI interaction as a coupled co-evolutionary system and by developing methods to measure, model, steer, and validate healthy co-adaptation, building on foundations established by my group in information-theoretic diagnostics [10], cognitive reasoning [48, 49, 50, 51], agent memory [52, 53], LLM personalisation [54, 55, 56], LLM meta-reasoning [57, 58], RL and reward hacking [47, 59, 60], and personalised AI tutoring [40, 61]. The proposed synthesis is not a collection of these components. It is a new closed learning loop in which **sparse grounded human capability becomes a trainable signal for long-horizon personalisation**. The project is high-risk because its core quantities, such as capability growth and human–AI coupling, are difficult to estimate from sparse longitudinal interaction data. Its transformative potential lies in establishing a new foundation for personalised AI, systems that optimise not for agreement, engagement, or short-term helpfulness, but for human capability growth.

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